11

Calculating with Conics

We [Kaplansky and Halmos] share a love of linear algebra. I think it is our conviction that we'll never understand infinite-dimensional operators until we have a decent mastery of finite matrices. And we share a philosophy about linear algebra: we think basis-free, we write basis-free, but when the chips are down we close the office door and compute with matrices like fury.

Irving Kaplansky
in Paul Halmos: Celebrating 50 Years of Mathematics

A major part of classical elementary geometry was concerned with the question of which constructions can be carried out with a straightedge and compass. The decisive primitive operations here are connecting two points by a line, drawing a circle with a radius given by two other points, and marking intersections of objects as new points. In our projective framework we do not have circles, but still can consider elementary constructions with the objects we have studied so far (points, lines, and conics). Since we are interested in particular in *calculating* with geometric objects, we in particular want to know how we can *compute* the results of geometric primitive operations if the parameters of the involved objects are given. So far, we can roughly associate the classical straightedge/compass operations with our projective operations in the following way:

- connecting two points by a line corresponds to the *join* operation and can be carried out by a cross product,
- intersecting two lines corresponds to the *meet* operation and is also carried out by the cross product,
- constructing a circle from two points can be associated to our construction of a *conic by five points* as described in Section 10.1.

Furthermore, we had additional operations for

- constructing polars of points and lines with respect to a conic (this included the calculation of a tangent),
- transforming points/lines/conics by projective transformations,
- calculating the matrix of the dual of a given conic.

So far we do not have an algebraic equivalent of the classical operations of intersecting a line and a circle and of intersecting two circles. This chapter is (among others) dedicated to this task. We will develop algebraic methods for intersecting conics with lines and conics with conics. Clearly, these operations can be carried out by solving corresponding systems of polynomial equations. However, we will try to make the calculations for these operations as natural as possible in the framework of homogeneous coordinates and matrix representations for conics. This chapter is meant as a collection of recipes for such kinds of primitive operations. Many of these recipes are used in the implementation of the dynamic geometry program Cinderella [112].

11.1 Splitting a Degenerate Conic

Before we turn our attention to the problem of intersecting a conic with other objects we will study how it is possible to derive homogeneous coordinates for the two lines of a degenerate conic from the matrix of a conic. This will turn out to be a useful operation later on.

Assume that a conic $\mathcal{C}_A$ is given by a symmetric matrix A such that as usual, $\mathcal{C}_A = \{p \mid p^T A p = 0\}$. If the conic is degenerate and consists of two lines or of one double line, then A will not have full rank. Thus we can determine a degenerate situation by testing $\det(A) = 0$. If a conic consists of two distinct lines with homogeneous coordinates g and h, then its symmetric (rank-2) matrix A can be written as $A = gh^T + hg^T$ (up to a scalar multiple). The matrices gh^T and hg^T would in principle generate the same conic, but they are not symmetric. However, knowing one of these matrices (for instance gh^T) would be equivalent to knowing the homogeneous coordinates of the two lines, since the columns of this matrix are just scalar multiples of g and the rows are just scalar multiples of h. Any nonzero column (resp. row) could serve as homogeneous coordinates for g (resp. h). This can be seen easily by observing that the disjunction of the conditions $\langle p, h\rangle = 0$ and $\langle p, g\rangle = 0$ can be written as

$$0 = \langle p, g\rangle \cdot \langle p, h\rangle = (p^T g)(h^T p) = p^T (gh^T) p.$$

So, splitting a degenerate conic into its two lines essentially corresponds to finding a rank-1 matrix B that generates the same conic as A. The quadratic form is linear in the corresponding matrix (i.e., we have $p^T(A + B)p = p^T Ap + p^T Bp$). Furthermore, those matrices for which the quadratic form

is identically zero are exactly the skew-symmetric matrices (those with $A^T = -A$). All in all, for decomposing a symmetric degenerate matrix A into two lines we have to find a skew-symmetric matrix B such that $A + B$ has rank 1. Thus in our case we have to find parameters λ, μ, and τ such that the following matrix sum has rank 1:

$$\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} + \begin{pmatrix} 0 & \tau & -\mu \\ -\tau & 0 & \lambda \\ \mu & -\lambda & 0 \end{pmatrix}.$$

The determinant of every 2×2 submatrix of a rank-1 matrix must vanish. Thus necessary conditions for the parameters are

$$\begin{vmatrix} a & b+\tau \\ b-\tau & c \end{vmatrix} = 0; \quad \begin{vmatrix} a & d-\mu \\ d+\mu & f \end{vmatrix} = 0; \quad \begin{vmatrix} c & e+\lambda \\ e-\lambda & f \end{vmatrix} = 0.$$

Resolving for λ, τ, and μ gives

$$\tau^2 = -\begin{vmatrix} a & b \\ b & c \end{vmatrix}; \quad \mu^2 = -\begin{vmatrix} a & d \\ d & f \end{vmatrix}; \quad \lambda^2 = -\begin{vmatrix} c & e \\ e & f \end{vmatrix}.$$

This determines the parameters λ, μ, and τ up to their sign. In principle one could test all eight possibilities to find a suitable choice that makes the entire matrix a rank-1 matrix. However, there is also a more direct way of calculating the values with their correct sign. For this we associate to the parameter vector $p = (\lambda, \mu, \tau)^T$ the skew-symmetric matrix

$$\mathcal{M}_p := \begin{pmatrix} 0 & \tau & -\mu \\ -\tau & 0 & \lambda \\ \mu & -\lambda & 0 \end{pmatrix}.$$

Left multiplication by the matrix $\mathcal{M}_p$ encodes performing a cross product with the vector p. A simple calculation shows that for any three-dimensional vector q we have

$$\mathcal{M}_p \cdot q = p \times q.$$

Lemma 11.1. *Let A be a rank-2 symmetric 3×3 matrix that defines a conic consisting of two distinct lines. Let p be the point of intersection of these lines. Then for a suitable factor α the matrix $A + \alpha\mathcal{M}_p$ has rank 1.*

Proof. Let g and h be homogeneous coordinates for the two lines. We may assume that these coordinates are scaled such that A has the form $gh^T + hg^T$. The intersection of these lines is $g \times h$. We have for a suitable factor α the equation $g \times h = \alpha p$. If we consider the difference $gh^T - hg^T$, we obtain the skew-symmetric matrix

$$gh^T - hg^T = \begin{pmatrix} 0 & g_1h_2 - g_2h_1 & g_1h_3 - g_3h_1 \\ g_2h_1 - g_1h_2 & 0 & g_2h_3 - g_3h_2 \\ g_3h_1 - g_1h_3 & g_3h_2 - g_2h_3 & 0 \end{pmatrix}.$$

Comparison of coefficients shows that this matrix is nothing other than $\mathcal{M}_{g\times h}$. Thus we obtain

$$gh^T - hg^T = \mathcal{M}_{g\times h} = \mathcal{M}_{\alpha p} = \alpha\mathcal{M}_p.$$

With this we obtain

$$A + \alpha\mathcal{M}_p = (gh^T + hg^T) + (gh^T - hg^T) = 2gh^T.$$

Thus the result must have the desired rank-1 form, and the theorem is proven. □

In particular, if for specific coordinates g and h we have $A = gh^T + hg^T$ and $p = g \times h$, we can choose the factor $\alpha = 1$ and obtain

$$A - \mathcal{M}_p = 2hg^T.$$

If we instead add the matrices, we obtain $A + \mathcal{M}_p = 2gh^T$. The previous lemma allows us to calculate the corresponding rank-1 matrix if a symmetric matrix of a degenerate conic is given. However, it has one big disadvantage: We know neither p nor α in advance. There are several circumstances (we will encounter them later) in which we, for instance, know p in advance. However, it is also possible to calculate p from the matrix A without too much effort. For this we have to use the formula

$$(gh^T + hg^T)^\triangle = -(g \times h)(g \times h)^T,$$

which we already used in Section 9.5. The matrix $B = (g \times h)(g \times h)^T$ is a rank-1 matrix pp^T with $p = g \times h$. Thus each row/column of this matrix is a scalar multiple of p. Furthermore, the diagonal entries of this matrix correspond to the squared coordinate values of $g \times h$. Thus one can extract the coordinates of $g \times h$ by searching for a nonzero diagonal entry of B, say $B_{i,i}$, and set $p = B_i/\sqrt{B_{i,i}}$, where B_i denotes the ith column of B. With this assignment we can calculate the matrix $A^\triangle + \mathcal{M}_p$, which gives either $2gh^T$ or $2hg^T$ depending on the sign of the square root. Altogether, we can summarize the procedure of splitting a matrix A that describes a conic consisting of two distinct lines.:

1: $B := A^\triangle$;

2: Let i be the index of a nonzero diagonal entry of B;

3: $\beta = \sqrt{B_{i,i}}$;

4: $p = B_i/\beta$, where B_i is the ith column of B;

5: $C = A + \mathcal{M}_p$;

6: Let (i, j) be the index of a nonzero element $C_{i,j}$ of C;

7: g is the ith row of C, h is the jth column of C.

After this calculation g and h contain the coordinates of the two lines. If A describes a conic consisting of a double line, then this procedure does not apply, since B will already be the zero matrix. Then one can directly split the matrix A by searching a nonzero row and a nonzero column.

11.2 The Necessity of "If" Operations

Compared to all our other geometric calculations (such as computing the *meet* of two lines, the *join* of two points, or a *conic through five* given points), the last computation for splitting a conic is considerably different. During the computation we had to inspect a 3×3 matrix for a nonzero entry in order to extract a nonzero row (or a nonzero column). One might ask whether it is possible to perform such a computation without such an inspection of the matrix, which intrinsically requires *branching* if one implements such a computation on a computer. Indeed, it is not possible to do the splitting operation for all instances without any branches. In other words, *there is no closed formula for extracting homogeneous coordinates for the two lines of a degenerate conic.* No matter which calculation is performed, there will always be sporadic special cases that are not covered by the concrete formula. The reason for this is essentially of a topological nature. No continuous formula can be used for performing the splitting operation without any exceptions. This is already the case for extracting the double line of a conic that consists of a double line, as the next theorem shows.

Theorem 11.1. *Let* $\mathrm{Deg} = \{ll^T \mid l \in \mathbb{R}^3 - \{(0,0,0)^T\}\}$ *be the set of all* 3×3 *symmetric rank-1 matrices. Let* $\phi\colon \mathrm{Deg} \to \mathbb{R}^3$ *be a continuous function that associates to each matrix* pp^T *a real scalar multiple* λp *of the vector* p*. Then there is a matrix* $A \in \mathrm{Deg}$ *for which* $\phi(A) = (0,0,0)^T$.

Proof. Assume that ϕ is a continuous function that associates to each matrix pp^T a scalar multiple λp of p. We consider the path $\tau : [0, \pi] \to \mathbb{R}^3$ with $\tau(t) = (\cos(t), \sin(t), 0)^T$. By our assumption, the function $\phi(\tau(t)\tau(t)^T)$ must be continuous on the interval $[0, \pi]$. We have $\tau(0) = (1,0,0)^T =: a$ and $\tau(\pi) = (-1,0,0)^T = -a$ and therefore $\phi(\tau(0)\tau(0)^T) = \phi(aa^T) = \phi((-a)(-a)^T) = \phi(\tau(\pi)\tau(\pi)^T)$. By the definition of ϕ we must have $\phi(\tau(t)\tau(t)^T) = \lambda(\tau(t))$. While t moves from 0 to π, the factor λ must itself behave continuously, since in every sufficiently small interval at least one of the coordinates of $\tau(t)$ is constantly nonzero. However, we have that $\phi(\tau(0)\tau(0)^T) = \tau(0)$ and

$\phi(\tau(\pi)\tau(\pi)^T) = \phi(\tau(0)) = -\tau(\pi)$. Since λ was assumed to be real this implies by the intermediate value theorem that for at least one parameter $t_0 \in [0, \pi]$ we must have $\lambda = 0$. □

The matrix ll^T that corresponds to the point p for which we have $\phi(ll^T) = (0,0,0)^T = 0 \cdot p$ does not lead to a meaningful nondegenerate evaluation of ϕ. The interpretation of this fact is a little subtle. It means that on the coordinate level there is no continuous way of extracting the double line of a conic $\mathcal{C}_A$ from a symmetric rank-1 matrix A. On the other hand, the considerations of the last sections show that on the level of geometric objects there is a way to extract the coordinates of the line l from the matrix ll^T which must be necessarily continuous in the topology of our geometric objects. There is just no way of doing these computations without using branching within the calculations. This was reflected by the fact that we explicitly had to search for nonzero entries in our 3×3 matrices.

In fact, the effect treated in this section is just the beginning of a long story that leads to the conclusion that elementary geometry and effects from complex function theory (such as monodromy, multivalued functions) are intimately interwoven. We will return to these issues in the very final section of this book.

11.3 Intersecting a Conic and a Line

After all this preparatory work the final task of this chapter turns out to be relatively simple. We want to calculate the intersection of a conic given by a symmetric 3×3 matrix A and a line given by its homogeneous coordinates l. Clearly the task is in essence nothing but solving a quadratic equation. However, we want to perform the operation that is as closely as possible related to the coordinate representation. For this we will use the operation of splitting a matrix that represents a degenerate conic as introduced in Section 11.1 as a basic building block. Essentially, the square root needed to solve a quadratic equation will be the one required for this operation.

Our aim will be to derive a closed formula for the degenerate conic that consists of the line l as double line and whose dual consists of the two points of intersection. Such a conic is given by a pair of matrices (A, B), where A is a rank-1 symmetric matrix and describes the double line l and where B is a symmetric rank-2 matrix with $B^\triangle = A$ that describes the dual conic and with this the position of the two points of intersection. Splitting the matrix B yields the two points of intersection.

So, how do we derive the matrix B? We will characterize the matrix via its properties. Let p and q be the two (not necessarily distinct) intersection points. The quadratic form $m^T B m$ must have the property that it vanishes for exactly those lines m that pass through (at least) one of the points p and q. A matrix with these properties is given by

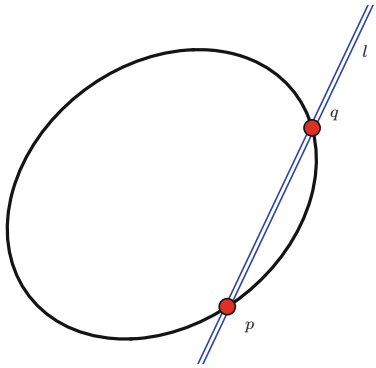

Fig. 11.1 Intersecting a conic and a line: Consider the line as a conic consisting of a double line and two points on it.

$$B = \mathcal{M}_l^T A \mathcal{M}_l.$$

To see this, we calculate the quadratic form $m^T Bm$. Using the property $\mathcal{M}_l m = l \times m$, we get

$$m^T Bm = m^T \mathcal{M}_l^T A \mathcal{M}_l m = (\mathcal{M}_l m)^T A(\mathcal{M}_l m) = (l \times m)^T A(l \times m).$$

The right side of this chain of equations can be interpreted as follows: $l \times m$ calculates the intersection of l and m. The condition $(l \times m)^T A(l \times m)$ tests whether this intersection is also on the conic $\mathcal{C}_A$. Thus as claimed, $\mathcal{M}_l^T A \mathcal{M}_l$ is the desired matrix B. It is the matrix of a dual conic describing two points on l. Splitting this matrix finally gives the intersections in question.

Compared to Section 11.1 we are this time in the dual situation. We want to split a matrix of a dual conic into two *points*. For this we have to transform it into an equivalent rank-1 matrix (one that defines the same conic) by adding a skew-symmetric matrix. For the splitting procedure we do not have to apply the full machinery of Section 11.1. This time we are in the good situation that we already know the skew-symmetric matrix up to a multiple. It must be the matrix $\mathcal{M}_l$, since l was by definition the join of the two intersection points. Thus the desired rank-1 matrix has the form

$$\mathcal{M}_l^T A \mathcal{M}_l + \alpha \mathcal{M}_l$$

for a suitably chosen factor α. This factor must be chosen in a way that the resulting matrix has rank 1. The parameter α can be simply determined by considering a suitable 2×2 submatrix of the resulting matrix.

All in all, the procedure of calculating the intersections of a line l and a conic given by A can be described as follows (without loss of generality we assume that the last coordinate entry of $l = (\lambda, \mu, \tau)^T$ is nonzero):

1: $B = \mathcal{M}_l^T A \mathcal{M}_l$;

2: $\alpha = \frac{1}{\tau}\sqrt{-\begin{vmatrix} B_{1,1}, & B_{1,2} \\ B_{1,2} & B_{2,2} \end{vmatrix}}$;

3: $C = B + \alpha \mathcal{M}_l$;

4: Let (i, j) be the index of a nonzero element $C_{i,j}$ of C;

5: p is the ith row of C, q is the jth column of C.

The choice of α in the second row ensures that the matrix C will have rank 1. The particular sign of α is irrelevant, since a sign change would result in interchanging the points p and q. If the entry τ of l were zero, one would have to take a different 2×2 matrix for determining the value of α. It is also a remarkable fact that if for some reason one is not interested in the individual coordinates of p and q but is interested in treating them as a pair, then all necessary information is already encoded in the matrix B. Furthermore, notice that for the calculation of the individual coordinates it is necessary to use a square-root operation just once. This is unavoidable, since intersecting a conic and a line can be used to solve a quadratic equation.

11.4 Intersecting Two Conics

Now, we want to intersect two conics. For this we will use the considerations of the last sections as auxiliary primitives and will reduce the problem of intersecting two conics to the problem of intersecting a conic with a line. Intersecting a conic and a line as considered in the previous section was essentially equivalent to solving a quadratic equation. Therefore it was necessary to use at least one square root operation. This will no longer be the case for the intersection of two conics. There the situation will be worse. Generally, two conics will have *four* more or less independent intersections. This indicates that it is necessary to solve a polynomial equation of degree four for the intersection operation. However, we will present a method that requires us to solve only a cubic (degree-3) equation. This results from the fact that in a certain sense the *algebraic difficulty* of solving degree-three and degree-four equations is essentially the same.

The idea for calculating the intersection of two conics is very simple. We assume that the two conics $\mathcal{C}_A$ and $\mathcal{C}_B$ are represented by matrices A and B. It is helpful for the following considerations to assume that the two conics have four real intersections. However, all calculations presented here can be carried out as well over the complex numbers. All linear combinations $\lambda A + \mu B$ of the matrices represent conics that pass through the same four points of intersection as the original matrices. In the *bundle of conics* $\{\lambda A + \mu B \mid \lambda, \mu \in \mathbb{R}\}$ we now search for suitable parameters λ and μ such that the matrix $\lambda A +$

μB is degenerate. After this we split the degenerate conic by the procedure described in Section 11.1. Then we just have to intersect the two resulting lines with one of the conics $\mathcal{C}_A$, $\mathcal{C}_B$ by the procedure described in Section 11.2.

In order to get a degenerate conic of the form $\lambda A + \mu B$ we must find λ, μ, such that

$$\det(\lambda A + \mu B) = 0.$$

At least one of the parameters λ, μ must be nonzero in order to get a proper conic. The problem of finding such parameters leads essentially to the problem of solving a cubic equation. To see this, one can simply expand the above determinant and observe that each summand contains a factor of the form $\lambda^i \mu^{3-i}$ with $i \in \{0, 1, 2, 3\}$. Collecting all these factors leads to a polynomial equation of the form

$$\alpha \cdot \lambda^3 + \beta \cdot \lambda^2 \mu + \gamma \cdot \lambda \mu^2 + \delta \cdot \mu^3 = 0.$$

We can easily calculate the parameters $\alpha, \beta, \gamma, \delta$ using the multilinearity of the determinant function. Assume that the matrix A consists of column vectors A_1, A_2, A_3 and that the matrix B consists of column vectors B_1, B_2, B_3. Expanding $\det(\lambda A + \mu B)$ yields

$$\begin{aligned}\det(\lambda A + \mu B) = \quad & \lambda^3 \, [A_1, A_2, A_3] \\ & + \lambda^2 \mu \, ([A_1, A_2, B_3] + [A_1, B_2, A_3] + [B_1, A_2, A_3]) \\ & + \lambda \mu^2 \, ([A_1, B_2, B_3] + [B_1, A_2, B_3] + [B_1, B_2, A_3]) \\ & + \quad \mu^3 \, [B_1, B_2, B_3].\end{aligned}$$

Thus we get

$$\begin{aligned}\alpha &= [A_1, A_2, A_3], \\ \beta &= [A_1, A_2, B_3] + [A_1, B_2, A_3] + [B_1, A_2, A_3], \\ \gamma &= [A_1, B_2, B_3] + [B_1, A_2, B_3] + [B_1, B_2, A_3], \\ \delta &= [B_1, B_2, B_3].\end{aligned}$$

If we find suitable λ, μ that solve $\alpha \cdot \lambda^3 + \beta \cdot \lambda^2 \mu + \gamma \cdot \lambda \mu^2 + \delta \cdot \mu^3 = 0$, then $\lambda A + \mu B$ will represent a degenerate conic. From this degenerate conic it is easy to calculate the intersections of the original conics. Thus the problem of intersecting two conics ultimately leads to the problem of solving a cubic polynomial equation (and this is unavoidable). The story of solving cubic equations goes back to the sixteenth century and has a long and exciting history. (This is one of *the* legends in mathematics, including human tragedies, challenges, vanity, competition. The main actors in this play were Scipione del Ferro, Anton Maria Fior, Nicolo Tartaglia, Girolamo Cardano, and Lodovico Ferrari, all of whom lived between 1465 and 1569. In Chapter 15 we will give a brief historical overview. We refer readers interested in elaborate details to the book by Yaglom [136], the novel "Der Rechenmeister" by Jörgensen [61], and the numerous articles on this topic on the Internet.)

For our purposes we will confine ourselves to a direct way for solving a cubic equation. Our procedure has three nice features compared to usual solutions of cubic equations. It works in the homogeneous setting where we ask for values of λ and μ instead of just a one-variable version that does not handle "infinite" cases properly. It needs to calculate exactly one square root and exactly one cube root, and we will not have to take care which specific roots of all complex possibilities we take. It works on the original cubic equation (most solutions presented in the literature work only for a reduced equation where $\beta = 0$).

All calculations in our procedure have to be carried out over the complex numbers, since it may happen that intermediate results are no longer real. For the computation we will need one of the cube roots of unity as a constant. We abbreviate it by

$$\omega = -\frac{1}{2} + i \cdot \sqrt{\frac{3}{4}}.$$

A reasonable procedure for solving the equation

$$\alpha \cdot \lambda^3 + \beta \cdot \lambda^2\mu + \gamma \cdot \lambda\mu^2 + \delta \cdot \mu^3 = 0$$

is given by the following sequence of operations (don't ask why the parameters and formulas work; it's a long story):

1: $W = -2\beta^3 + 9\alpha\beta\gamma - 27\alpha^2\delta$;

2: $D = -\beta^2\gamma^2 + 4\alpha\gamma^3 + 4\beta^3\delta - 18\alpha\beta\gamma\delta + 27\alpha^2\delta^3$;

3: $Q = W - \alpha\sqrt{27D}$;

4: $R = \sqrt[3]{4Q}$;

5: $L = (2\beta^2 - 6\alpha\gamma, -\beta, R)^T$;

6: $M = 3\alpha(R, 1, 2)^T$;

The two vectors L and M are the key to finding the three solutions for the final computation of λ and μ. For this we have to compute

$$\begin{pmatrix} \omega & 1 & \omega^2 \\ 1 & 1 & 1 \\ \omega^2 & 1 & \omega \end{pmatrix} \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix}, \qquad \begin{pmatrix} \omega & 1 & \omega^2 \\ 1 & 1 & 1 \\ \omega^2 & 1 & \omega \end{pmatrix} \begin{pmatrix} M_1 \\ M_2 \\ M_3 \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}.$$

The pairs (λ_1, μ_1), (λ_2, μ_2), and (λ_3, μ_3) are the three solutions of the cubic equation. In Step 4 of the above procedure we have to choose a specific cube root. If R is one cube root, the other two cube roots are ωR and $\omega^2 R$. The reader is invited to convince himself that no matter which of these cube roots we take, we obtain the same set of solutions (in fact they are permuted). Similarly, if we change the sign of the square root in Step 3, two of the solutions are interchanged.

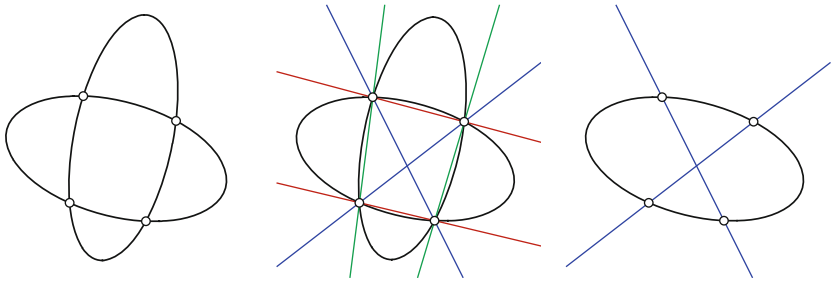

Fig. 11.2 Intersecting two conics: The original problem—the three degenerate conics—the reduced problem.

After collecting all these pieces it is easy to use them to give a procedure for intersecting two conics. We just have to put the pieces together. If A and B are the two matrices representing the conics, we can proceed in the following way (this time we just give rough explanations of the steps instead of detailed formulas).

1: Calculate $\alpha, \beta, \gamma, \delta$ as described before.

2: Find a solution (λ, μ) of the cubic equation $\alpha \cdot \lambda^3 + \beta \cdot \lambda^2 \mu + \gamma \cdot \lambda \mu^2 + \delta \cdot \mu^3 = 0$.

3: Let $C = \lambda A + \mu B$.

4: Split the conic C into two lines g and h.

5: Intersect both lines g and h with the conic $\mathcal{C}_A$.

All in all, we obtain four intersections, two for each of the two lines g and h. Figure 11.2 illustrates the process. We start with the two conics and calculate one degenerate conic that may be obtained as a linear combination of them. Then we split this degenerate conic into two lines and intersect each of these lines with one of the original conics. There are a few subtleties concerning which intermediate results are real and which are complex. We will discuss them in the next section.

11.5 The Role of Complex Numbers

Let us take a step back and look at what we did in the previous section. At the beginning of our discussions on intersecting conics with conics we said that all calculations should be carried out over the complex numbers. In all previous chapters we focused on *real* projective geometry. However, solving the cubic equation may make it inevitable to have complex numbers at least

as an intermediate result (if the value of D calculated in step two of our cubic equation procedure becomes negative).

We first discuss what it means to have a *real* object in the framework of homogeneous coordinates. In all our considerations so far we have agreed to identify coordinate vectors that differ only by a scalar multiple. If we work with complex coordinates we will do essentially the same. However, this time we will also allow complex multiples. We will call a coordinate vector p "real" if there is a (perhaps complex) scalar s such that $s{\cdot}p$ has only real coordinates. In this sense the vector $(1+2i, 3+6i, -2-4i)^T$ represents a real object, since we can divide by $1+2i$ and obtain the real vector $(1, 3, -2)^T$. Similarly, the vector $(1, i, 0)$ is a proper complex vector, since no matter by which nonzero number we multiply it, at least one of the entries will be complex. It is an amazing fact that we may get real objects from calculations with proper complex objects. Consider the following calculation, which could be considered the join of two proper complex points:

$$\begin{pmatrix} 1+2i \\ 3+i \\ 1-i \end{pmatrix} \times \begin{pmatrix} 1-2i \\ 3-i \\ 1+i \end{pmatrix} = \begin{pmatrix} 8i \\ -6i \\ 10i \end{pmatrix} = 2i \cdot \begin{pmatrix} 4 \\ -3 \\ 5 \end{pmatrix}.$$

The coordinates of the result are complex, but they still represent the real object $(4, -3, 5)^T$. The reason for this is that in this example we took the join of two complex conjugate points. More generally, we get for a point $p + iq$, with real vectors p and q,

$$(p+iq) \times (p-iq) = p \times p + (iq) \times p + p \times (-iq) + (iq) \times (-iq) = 2i(q \times p).$$

The join (meet) of two conjugate complex points (lines) is a real geometric object. This perfectly fits in our geometric intuition. Imagine you intersect a line l with a conic $\mathcal{C}_A$. If the line is entirely outside the conic, they do not have real intersections. As algebraic solution we get two complex conjugate points. Joining these two points, we get the real line l again. Generally, when we deal with homogeneous coordinates we will call an object real if there is a scalar multiple that simultaneously makes all entries real. We will apply this definition to points, lines, conics, transformations and also to bundles of objects parameterized by homogeneous coordinates.

Now let us return to the problem of intersecting conics. If we have two real conics $\mathcal{C}_A$ and $\mathcal{C}_B$, then the parameters $\alpha, \ldots, \delta$ for the cubic equation will be real as well. A cubic equation in $\mathbb{C}$ has three solutions (if necessary counted with multiplicity). Except for degenerate cases in which two or all three of these solutions coincide, we will have one of the following two cases: Either all these three solutions are real or just one of the solutions is real and the other two are complex conjugates. In any case we will have at least one real solution. The term "real solution" in the homogeneous setup in which we identify scalar multiples of (λ, μ) means that there is a scalar s that makes

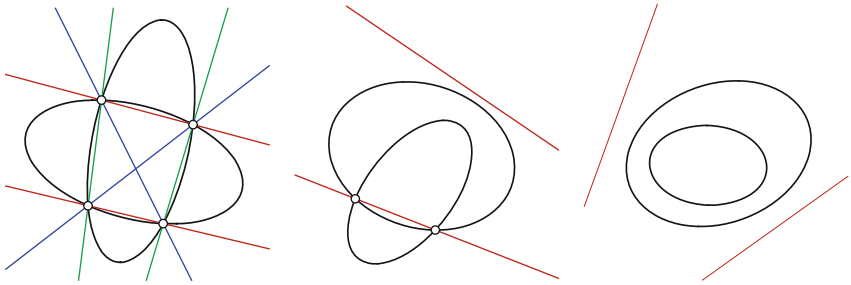

Fig. 11.3 Real degenerate conics from a pair of conics.

both coordinates of $s \cdot (\lambda, \mu)$ simultaneously real. In other words, the quotient λ/μ is real. Such a real solution corresponds to the fact that the degenerate conic $C = \lambda A + \mu B$ is real again. If the conics have four points in common, then we will have indeed three real solutions corresponding to the three real solutions of the cubic equation. If the two conics have only two points in common, then there will be only one real solution. This solution will consist of two lines. One of these lines is the join of the intersection points; the other is another line not hitting the conics in any real points, but still it passes through the other two intersections of the conics, which are properly complex and conjugates of each other. If the two conics do not have any real intersection, then there is still a real solution to the cubic equation. We still get one real degenerate conic in the bundle generated by the two conics. This degenerate conic consists of two lines each of them passing through a pair of complex conjugate intersection points. The three situations are shown in Figure 11.3.

It is also very illuminating to study the case in which we pass from the "three real solutions" situation to the "one real two complex conjugates" situation. In this case the cubic equation will have a real double root and a real single root. This means that two of the three real degenerate conics in the bundle $\lambda A + \mu B$ will coincide. Geometrically, this corresponds to the case in which the two conics meet tangentially at one point and have two real intersections elsewhere. Figure 11.4 shows a situation an "epsilon" before the tangent situation. In the picture we still have four real intersections. However, two of them approach each other tightly. It can be seen how in this case two of the degenerate conics almost coincide as well, and how one of the lines of the other conic almost becomes a tangent. In the limit case the two first degenerate conics will really coincide and the tangent line of the third conic will really be a tangent at the point of tangency of the two conics.

What does all this mean for our problem of calculating the intersections of two conics? If we want to restrict ourselves to real calculations whenever possible, then it might be reasonable to pick the real solution of the cubic and

proceed with this one. Then we finally have to intersect two real lines with a conic. Still it may happen that one or both of the lines do not intersect the conic in real points. However, whenever we have real intersection points we will find them by this procedure by intersecting a real line and a real conic.

If we think of implementing such an operation in a computer program, it may also be the case that the underlying math library can properly deal with complex numbers (it should anyway for solving the cubic equation). In such a case we do not have to explicitly pick a real solution. We can take any solution we want. If we by accident pick a complex solution, then we will get a complex degenerate conic, which splits into two complex lines. However, intersecting the lines with one of the original conics will result in the correct intersections. If the correct intersections turn out to be real points, then it just may be necessary to extract a common complex factor from the homogeneous coordinates.

11.6 One Tangent and Four Points

As a final example of a computation we want to derive a procedure that calculates a conic that passes through four points and at the same time is tangent to a line (see Figure 11.5 for the two possible solutions for an instance of such a problem). Based on the methods we have developed so far, there are several ways to approach this construction problem. Perhaps the most straightforward is to construct a fifth point on the conic and then

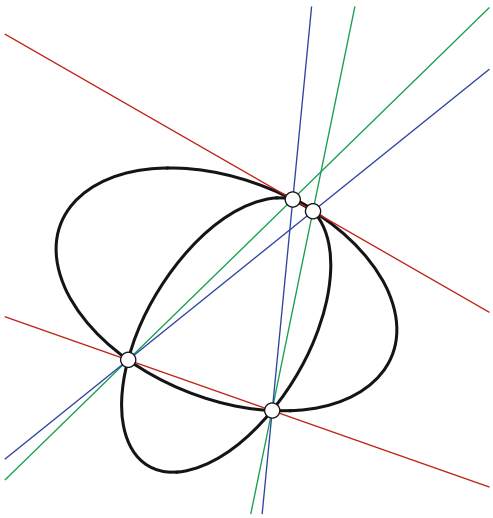

Fig. 11.4 An almost degenerate situation.

calculate the conic through these five points. This fifth point may be chosen on the line itself. Then it must be at the position where the resulting conic touches the line. Generally, there are two possible positions for such a point, corresponding to the two possible conics that satisfy the tangency conditions.

The crucial observation that allows us to calculate these two points is the fact that pairs of points that are generated by intersecting all possible conics through four given points with a line l are the pairs of points of a projective involution on the line. For this, recall that the conics through four points A, B, C, D have the form

$$\mathcal{C}_\lambda := \{p \mid (A, B; C, D)_p = \lambda\},$$

where λ may take an arbitrary value in $\mathbb{R} \cup \{\infty\}$. For each such conic there exists a pair of points on l that are as well on $\mathcal{C}_\lambda$. These pairs of points may be both real, both complex, or coinciding. If the points coincide, then we are exactly in the tangent situation. Let $\{p_\lambda, q_\lambda\}$ be the pair of such points on the conic $\mathcal{C}_\lambda$. Then we have the following:

Lemma 11.2. *There is a projective involution τ on l such that for any $\lambda \in \mathbb{R} \cup \{\infty\}$ we have $\tau(p_\lambda) = q_\lambda$.*

Proof. Instead of giving an algebraic proof we will directly use a geometric argument. With the help of Pascal's theorem we can construct the point q_λ from A, B, C, D, and p_λ without explicit knowledge of the conic $\mathcal{C}_\lambda$. Figure 11.6 shows the construction. The black and white elements of the picture depend only on A, B, C, D and p_λ. Starting with the point p_λ we have to

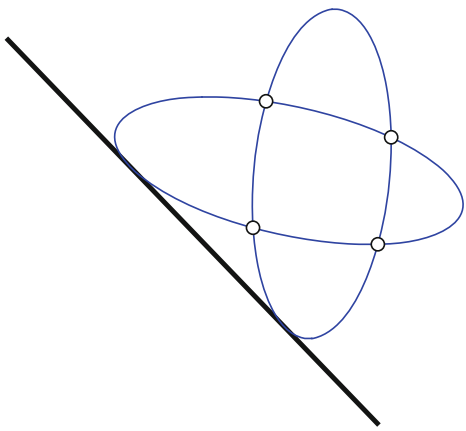

Fig. 11.5 Conics through four points and tangent to a line.

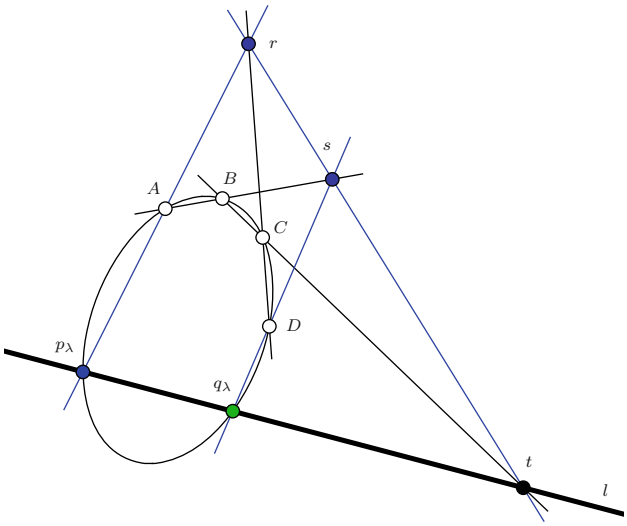

Fig. 11.6 Using Pascal's theorem to construct the second intersection.

construct first point r by intersecting $\overline{p_\lambda A}$ with $\overline{CD}$, then we construct s by intersecting $\overline{rt}$ with $\overline{AB}$. Finally, we derive point q_λ by intersecting $\overline{sD}$ with l. In homogeneous coordinates the whole sequence of construction steps can be expressed as a sequence of cross-product operations that use the coordinates of point p_λ exactly once. Thus if a and b are homogeneous coordinates of two arbitrary distinct points on l and we express a point p on l by $p = \alpha a + \beta b$, then a sequence of operations

$$x_k \times (x_{k-1} \ldots \times (x_2 \times (x_1 \times (\alpha a + \beta b))) \ldots) =: \alpha' a + \beta' b$$

can be expressed as a single matrix multiplication

$$A \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}.$$

Thus it is a projective transformation by the fundamental theorem of projective geometry. It is clearly an involution, since the same construction could be used to derive p_λ from q_λ. □

Remark 11.1. Alternatively, the proof could also be carried out in purely algebraic terms. We briefly sketch this. Let again a and b be two homogeneous coordinates of two arbitrary distinct points on l and express a point p on l as $p = \alpha a + \beta b$. The two points on l on the conic $\mathcal{C}_\lambda$ satisfy the relation

$$[\alpha a + \beta b, A, C][\alpha a + \beta b, B, D] = \lambda[\alpha a + \beta b, A, D][\alpha a + \beta b, B, C].$$

Resolving for α and β yields the equation

$$(\alpha, \beta)(X + \lambda Y)\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = 0,$$

where X and Y are suitable symmetric 2×2 matrices in which all parameters of the first equation have been encoded. Just knowing X and Y is enough information to derive the matrix A with the property $Ap_\lambda = q_\lambda$. The matrix A can be calculated by the amazingly simple formula

$$A = X^\triangle Y - Y^\triangle X.$$

The reader is invited to check algebraically that A is an involution and that it converts one solution of the quadratic equation into the other.

Lemma 11.2 reveals another remarkable connection between conics and quadrilateral sets. If we consider three different conics through four points A, B, C, D and consider the three point pairs that arise from intersecting these conics with line l, then these three pairs of points form a quadrilateral set. This is a direct consequence of Lemma 11.2 and Theorem 8.4 that connects projective involutions to quadrilateral sets. A corresponding picture that illustrates this fact is given in Figure 11.7. The incidence structure that is supported by the four black points and their joins is a witness that the six points on the black line form a quadrilateral set. In fact, the six lines of this witness construction can themselves be considered three degenerate conics that intersect the black line in a quadrilateral set.

Now we have collected all necessary pieces to construct the two tangent conics to l through A, B, C, D. The four points induce a projective involution τ on l that associates the pairs of points that arise by intersection with a conic through A, B, C, D. What we are looking for in order to construct a tangent conic are the two fixed points of the involution τ. Our considerations after Theorem 8.4 in Section 8.6 showed that these two fixed points x and y are simultaneously harmonic to all point pairs $(p, \tau(p))$. Thus we can reconstruct the position of these points if we know two such point pairs by solving a quadratic equation. We can construct even three such point pairs by considering the three degenerate conics through A, B, C, D. The situation is illustrated in Figure 11.8. Considering the line l as $\mathbb{RP}^1$ and working with homogeneous coordinates on this space, x and y must satisfy the equations

$$[a_1, x][a_2, y] = -[a_1, y][a_2, x] \text{ and } [b_1, x][b_2, y] = -[b_1, y][b_2, x].$$

These equations are solved by the two solutions

$$x = \sqrt{[a_2, b_1][a_2, b_2]}a_1 + \sqrt{[a_1, b_1][a_1, b_2]}a_2,$$
$$y = \sqrt{[a_2, b_1][a_2, b_2]}a_1 - \sqrt{[a_1, b_1][a_1, b_2]}a_2$$

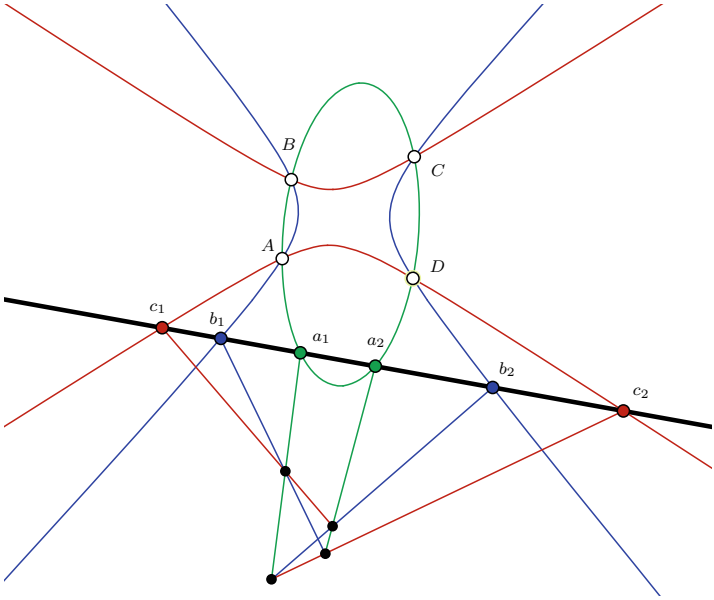

Fig. 11.7 Quadrilateral sets from bundles of conics.

(observe the beautiful symmetry of the solution). This solution can be derived as a variant of Plücker's μ technique if we try to express the solution as a linear combination $\lambda a_1 + \mu a_2$. The solution can be easily verified by plugging the expressions for x and y into the two equations and expanding the terms.

All in all, the procedure of calculating a conic through A, B, C, D tangent to l can be summarized as follows (we formulate the procedure so that the transition to the coordinates on l is only implicitly used):

1: Construct the four intersections $a_1 = \overline{AC} \wedge l$, $a_2 = \overline{BD} \wedge l$, $b_1 = \overline{AB} \wedge l$, $b_2 = \overline{CD} \wedge l$.

2: Choose an arbitrary point o not on l.

3: Let $x = \sqrt{[o, a_2, b_1][o, a_2, b_2]}a_1 + \sqrt{[o, a_1, b_1][o, a_1, b_2]}a_2$.

4: Let $y = \sqrt{[o, a_2, b_1][o, a_2, b_2]}a_1 - \sqrt{[o, a_1, b_1][o, a_1, b_2]}a_2$.

5: Return the two conics through A, B, C, D, x and through A, B, C, D, y.

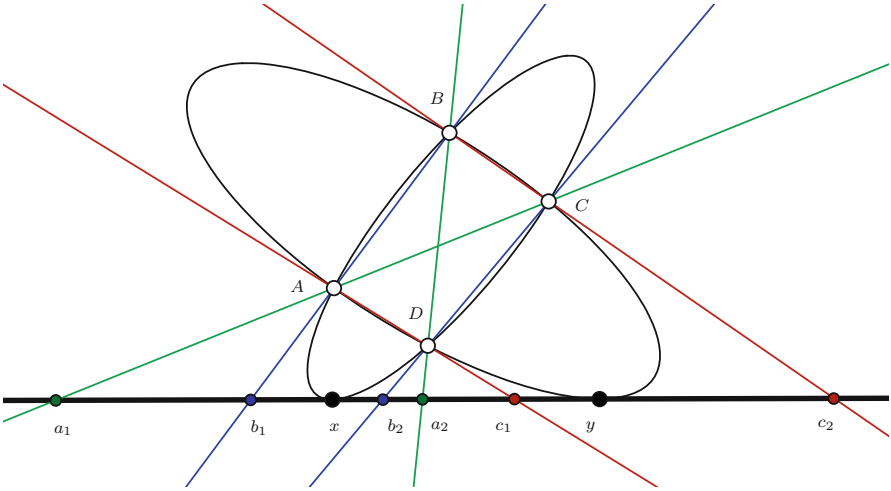

Fig. 11.8 The final construction.